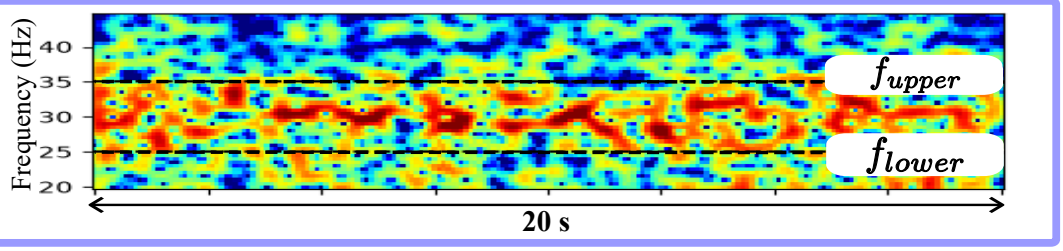
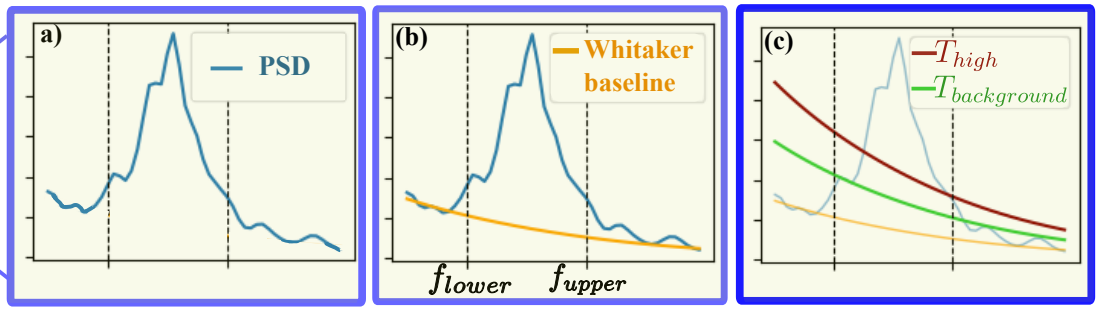
Input

- Spectrogram
- frequency interval
$I_f = [f_{lower}, f_{upper}]$



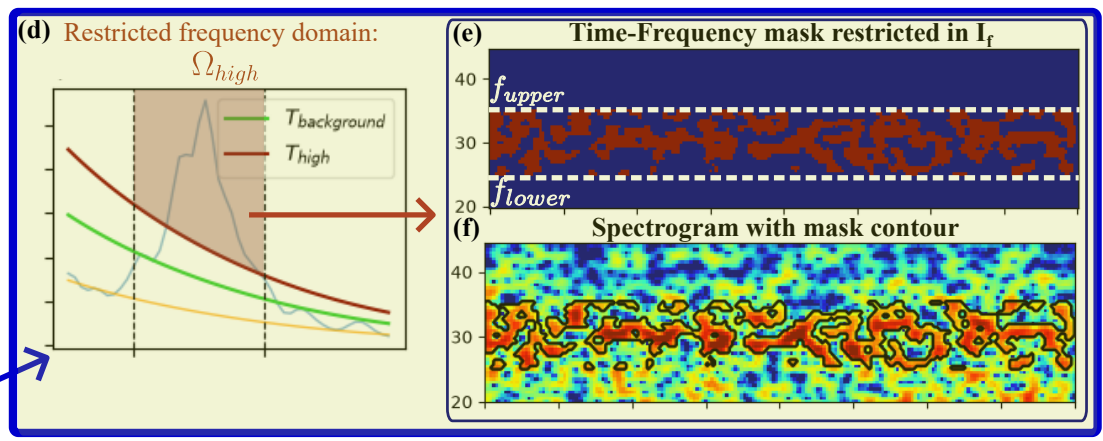
Step 1: Dynamic PSD thresholding

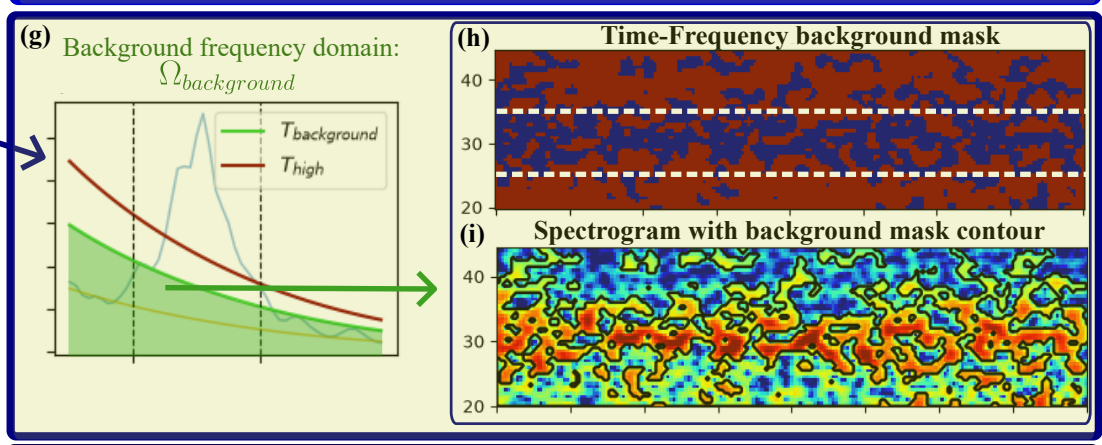
- computing PSD
- Whitaker baseline B fitting
- Thresholds scaling:
$T_{high}(f) = a_{high}B(f)$
$T_{background}(f) = a_{background}B(f)$



Step 2: Intensity pixel segmentation

- mask for pixels within I_f above T_{high}
- mask for pixels below $T_{background}$





Step 3: Mask expansion

- Expansion using Watershed algorithm

